Design regret in clustered wind farms: how wind rose directionality and optimization thoroughness shape the cost of ignoring neighbors

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Abstract. When a wind farm is designed in isolation—ignoring the wakes of potential neighboring farms—how much energy is sacrificed? We define this loss as *design regret*: the difference in annual energy production between a layout re-optimized with knowledge of neighbors and the original isolation-designed layout, both evaluated under the same neighbor wakes. Using greedy adversarial neighbor placement and converged multistart optimization ($K = 500$ random restarts), we sweep the two-parameter elliptical wind rose space of Hart (2025) and find that design regret varies six-fold across wind climate: from 17 GWh/yr (0.4 % of AEP) under bidirectional conditions to 101 GWh/yr (2.5 %) under concentrated unidirectional conditions. Wind rose directionality—specifically the folding parameter controlling uni- versus bidirectionality—is the dominant driver. Regret cross-sections at $K = 500$ confirm an "ambush effect": peak vulnerability occurs at oblique angles to the dominant wind direction, not along it. A self-interested neighbor comparison reveals that under concentrated unidirectional wind, a neighbor optimizing its own production causes nearly double the design regret of a fixed reference farm, because it concentrates turbines in the high-wind upwind corridor. Critically, we find that the computational effort required for reliable regret estimation far exceeds standard practice: five random starts capture only 55–86 % of the converged regret. We recommend a minimum of 300 multistart optimizations for any study computing AEP differences between design strategies. Buffer distance analysis shows that regret decay depends on wind rose shape: concentrated unidirectional sites require buffers exceeding $40D$ (9.6 km), while bidirectional sites need minimal separation.

1 Introduction

The North Sea is entering an era of unprecedented wind farm clustering. Denmark's Vindmølleparken energy island will connect up to 10 GW across multiple farms in a shared concession zone; the German Bight already hosts over 8 GW with plans for 30 GW by 2050; and the broader North Sea basin is projected to reach 300 GW as part of Europe's 500 GW offshore target. At this density, wind farms no longer operate in isolation. Cluster wake losses in the German Bight are projected to reduce capacity factors by 30 %, with half of the loss attributable to cross-border wake accumulation (Fliegner et al., 2025). The phenomenon—increasingly termed "wind theft" (Finseraas et al., 2024; Du et al., 2026)—has prompted the UK to issue the first national policy guidance on inter-farm wake effects and NYSERDA to recommend minimum inter-farm spacing of 4 nmi (NYSERDA, 2024). Wake-induced velocity deficits have been measured at distances exceeding 50 km (Schneemann

et al., 2020; Canadillas et al., 2022), and inter-farm losses are projected to exceed the impact of a century of climate change on North Sea wind resources (Warder and Piggott, 2025).

Yet these emerging policy responses share a common gap: they treat all sites uniformly. Buffer distance recommendations, "good neighbour" design principles, and compensation frameworks do not account for variation in wind climate. This is a significant omission. A site where wind blows predominantly from one direction creates a single, predictable wake corridor that a neighbor-aware layout can avoid. A site with bidirectional wind creates competing corridors that no single rearrangement can escape. The value of coordinated design—and the cost of failing to coordinate—should depend fundamentally on the directional structure of the wind resource.

We formalize this intuition through the concept of *design regret*: the AEP difference between a layout designed with perfect neighbor foresight and one designed in isolation, both evaluated under the same neighbor wakes:

$$R = \mathrm{AEP}(\mathbf{x}^*_{\mathrm{cons}}, \mathbf{n}) - \mathrm{AEP}(\mathbf{x}^*_{\mathrm{lib}}, \mathbf{n}). \tag{1}$$

Design regret isolates the cost of the design decision itself, separate from the total AEP reduction caused by neighbor wakes. By sweeping the elliptical wind rose parameterization of Hart (2025)—which separates directional concentration from bidirectionality—we quantify how wind rose shape governs design regret. The answer is dramatic: regret varies six-fold across the shape space, from 17 GWh/yr under bidirectional conditions to 101 GWh/yr under concentrated unidirectional conditions. Buffer distance requirements inherit this dependence.

Quick et al. (2026) applied this framework to the Danish Energy Island cluster and found negligible design regret for all nine neighbor configurations. We show this is not a general property of offshore wind, but a consequence of DEI's partially bidirectional wind climate ($f \approx 0.38$). Sites in the southern North Sea with strongly unidirectional southwesterly winds sit in the high-regret region of the shape space and are substantially more vulnerable to uncoordinated cluster development.

However, accurately computing these quantities reveals a methodological challenge that undermines all existing regret-like calculations. Layout optimization is highly multimodal (Thomas et al., 2023; Valotta Rodrigues et al., 2024), and regret—as the *difference* between two optimization outcomes—amplifies the effect of local optima. We show that five random starts, more than many published studies employ, capture only 55–86 % of the converged regret.

This paper makes four contributions:

1. A convergence study demonstrating that 300+ multistart optimizations are needed for reliable regret estimation—over an order of magnitude more than typical practice.

2. A systematic sweep of the wind rose shape space showing that design regret varies six-fold with directionality, with concentrated unidirectional sites most vulnerable.

3. Buffer distance analysis quantifying how neighbor separation reduces regret, and how the decay rate depends on wind rose shape.

4. Regret cross-sections revealing an "ambush effect" in which peak design vulnerability occurs at oblique angles to the dominant wind direction.

2 Methods

2.1 Problem formulation

We consider a target wind farm of $N_t = 50$ IEA 15 MW turbines ($D = 240\,m$, hub height 150 m) within a fixed polygonal boundary derived from the Danish Energy Island concession zone. Minimum inter-turbine spacing is $4D$ (960 m). This configuration is representative of near-future North Sea developments: the IEA 15 MW turbine is the reference class for planned GW-scale projects, and the DEI polygon provides a realistic, irregular boundary.

The liberal layout $\mathbf{x}^*_{\text{lib}}$ maximizes AEP assuming no external turbines. The conservative layout $\mathbf{x}^*_{\text{cons}}$ maximizes AEP with specified neighbors $\mathbf{n}$. Both are evaluated with neighbors present to compute regret via Eq. (1). By construction, $R \geq 0$ when both reach global optima. We enforce this computationally through *regret pooling*: the conservative AEP is taken as $\max(\text{best conservative AEP},\ \text{liberal AEP with neighbors})$, guaranteeing non-negative regret even with imperfect optimization.

2.2 Wake model

We use the Bastankhah and Porté-Agel (2014) Gaussian deficit model with $k = 0.04$ and root-sum-of-squares superposition. The simulation is implemented in JAX (Bradbury et al., 2018), enabling automatic differentiation through the wake model. AEP is computed over 24 equally-spaced wind direction sectors at a constant wind speed of 9 m/s:

$$\text{AEP} = 8760 \sum_{k=1}^{24} w_k \sum_{i=1}^{N_t} P_i(U_{i,k}), \tag{2}$$

where w_k is the sector frequency, P_i is the turbine power curve, and $U_{i,k}$ is the effective wind speed at turbine i under direction k.

2.3 Wind rose parameterization

Wind rose shape is parameterized using the elliptical model of Hart (2025), in which the directional frequency distribution is derived from the projection of a unit circle onto an ellipse. The sector frequency at direction θ is proportional to the projected length of the ellipse along θ, rotated to align with the prevailing direction θ_{prev} and folded to control bidirectionality. Two shape parameters control the distribution:

- **Shape parameter** $a > 0$: controls directional concentration. The special value $a = 1/\sqrt{\pi} \approx 0.564$ recovers a uniform distribution. Larger a produces a more concentrated rose with a narrower peak around the prevailing direction.
- **Folding parameter** $f \in [0, 1]$: controls symmetry between opposing directions. At $f = 0$, the distribution is bidirectional—equal probability of wind from the prevailing direction and its opposite. At $f = 1$, it is fully unidirectional.

This parameterization separates two physically distinct wind climate properties: how concentrated the energy is in direction (controlled by a) and whether a secondary opposing mode exists (controlled by f). The prevailing direction is fixed at $\theta_{\text{prev}} = 270°$ and wind speed at 9 m/s across 24 equally-spaced sectors. We sweep $a \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$ and

$f \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ (35 combinations). This range spans the variety of North Sea wind climates, from the strongly unidirectional southwesterlies of the German Bight to the more bidirectional conditions at the Danish Energy Island.

Fitting the DEI wind rose with the elliptical model yields $a \approx 0.64$, $f \approx 0.38$ ($R^2 = 0.84$), placing it in the moderate-concentration, partially-bidirectional regime.

2.4 Adversarial neighbor placement: upper bound on regret

To establish a worst-case upper bound on design regret, we place individual neighbor turbines using a greedy grid search. A candidate grid with $5D$ spacing extends $50D$ from the target boundary ($\sim$1567 candidates). At each of 30 greedy steps, the single candidate position maximizing incremental regret is selected, each requiring a full K-start conservative optimization.

This adversarial approach places unconstrained individual turbines—not realistic wind farms—and is designed to find the *maximum achievable regret* for a given wind rose. It provides an upper bound because: (i) individual turbines operate at full thrust coefficient, whereas turbines within a real neighboring farm would experience internal wake losses; and (ii) the greedy algorithm optimizes placement purely to damage the target, whereas a real developer would optimize for their own production. The results from this analysis should be interpreted as the ceiling on design regret, not as a prediction of what a realistic neighbor would cause.

2.5 Regret cross-sections: realistic neighbor scenario

To assess design regret under more realistic conditions, we introduce the *regret cross-section*. An identical copy of the target farm—with the same number of turbines, boundary polygon, and liberal-optimized layout ($K_{\text{lib}} = 300$)—is placed at each combination of 24 bearings (15° steps) and 7 buffer distances (2, 5, 10, 15, 20, 30, 40D), producing 168 evaluations per wind rose case (Fig. 1).

2.5.1 Inter-farm distance definition.

Buffer distance is defined as the minimum Euclidean distance between the boundary polygons of the target and reference farms (not centroid-to-centroid). For each (bearing, distance) pair, we iteratively compute the centroid offset required to achieve the specified boundary gap, accounting for the irregular polygon shape. At each configuration, we also record the minimum turbine-to-turbine distance between the two farms. This boundary-gap definition ensures that no turbines from the reference farm overlap the target farm area at any bearing or distance, which is critical for physically meaningful results.

2.5.2 Reference farm construction.

The reference farm is an identical translated copy of the target: same boundary polygon, same number of turbines ($N_t = 50$), and same liberal-optimized layout. This represents the most natural "what-if" scenario: a neighboring developer builds the same farm design nearby.

Table 1. Design regret (GWh/yr) at selected K values.

Configuration	K=5	K=100	K=300	K=500	K=2000	K_5/K_{2000}
Conc. unidir. ($5D$)	15.9	15.9	26.2	26.2	26.6	60%
Conc. unidir. ($20D$)	67.9	67.9	79.3	79.3	79.3	86%
Mod. bidir. ($5D$)	24.1	43.1	43.6	43.6	43.6	55%

At each position, the target farm is re-optimized with $K = 300$ restarts (matching K_{lib}) while the reference farm remains fixed. The resulting polar map separates bearing and distance effects and reveals whether vulnerability is isotropic or directionally concentrated.

2.6 Multistart optimization and convergence

Each layout optimization uses gradient-based stochastic gradient descent (SGD) (Quick et al., 2023) with a constant learning rate phase followed by a decay phase. The convergence study (Section 3.1) sweeps iteration counts from 100 to 10000 at $K = 500$ starts and finds that regret stabilizes by 2000 total iterations (within 5% of the 10000-iteration value for all configurations tested). Production cross-section runs therefore use 2000 iterations; the greedy grid sweep uses 5000 iterations for additional margin.

Restarts include grid initialization, the liberal layout (ensuring regret pooling), and random polygon-sampled positions. The liberal layout is included as a conservative candidate to guarantee non-negative regret by construction. Following the convergence analysis of Quick et al. (2026), who found ∼400 starts needed for 66-turbine liberal layout convergence, we perform a dedicated convergence study for the regret computation (Section 3.1).

Production configurations: $K = 500$ for the greedy grid sweep and buffer analysis (upper bound); $K_{\text{lib}} = K_{\text{cons}} = 300$ for cross-sections (realistic scenario), with equal K for liberal and conservative to avoid optimization asymmetry artifacts.

3 Results

3.1 Convergence study: how many starts are enough?

We sweep K from 1 to 2000 for three configurations spanning the wind rose space (Table 1, Fig. 2). A 5×5 reference neighbor farm is placed at a fixed bearing and distance for each.

At $K = 5$, regret captures only 55–86 % of the converged value. The moderate bidirectional case is worst: $K = 5$ finds only 55 % of true regret because the landscape has many local optima with similar standalone AEP but different neighbor-present AEP. All configurations stabilize by $K = 300$. SGD iteration convergence is much faster: at $K = 500$, regret at 5000 iterations is within 3 % of the 10000-iteration value.

The root cause is catastrophic cancellation: regret is a *difference* between two large AEP values, so a 0.5 % error in either optimization can produce a 50 % error in regret. This applies to any study computing AEP differences between strategies—robust optimization, Pareto analysis, or inter-farm coordination assessments.

3.2 Wind rose shape controls design regret

Figure 3 shows design regret across the (a, f) space at $K = 500$. Regret ranges from 17.2 GWh (0.4 % of AEP) at $a = 0.5$, $f = 0.0$ to 100.6 GWh (2.5 %) at $a = 0.9$, $f = 1.0$—a six-fold range.

The folding parameter f is the dominant factor (Fig. 4). Mean regret increases monotonically from 45.2 GWh at $f = 0$ to 77.9 GWh at $f = 1$. Concentration (a) has a weaker, non-monotonic effect.

Unidirectional wind ($f = 1$) creates a single wake corridor that the liberal layout is blind to; the conservative optimizer shifts turbines laterally to escape it. Under bidirectional conditions ($f = 0$), wakes arrive from opposing directions and no lateral shift helps—regret is inherently limited.

For North Sea planning, this means that sites with strongly unidirectional southwesterly winds (characteristic of the German Bight and southern North Sea) face higher design regret than sites with more bidirectional conditions (such as Danish waters). The DEI wind rose ($a \approx 0.64$, $f \approx 0.38$) falls in the moderate-regret region, consistent with the negligible tradeoffs found by Quick et al. (2026).

3.3 Buffer distance modulates regret

Figure 5 shows regret vs. buffer distance at $K = 500$ for three wind roses.

For concentrated unidirectional wind ($a = 0.9$, $f = 1.0$), regret drops from 2.5 % at $2D$ to 0.4 % at $40D$ (9.6 km). The bidirectional case ($a = 0.5$, $f = 0.0$) shows gentle decay from 0.4 % to 0.2 %.

The NYSERDA 4 nmi ($\sim 31D$) recommendation reduces regret to <1 % for all wind roses tested, but is unnecessarily conservative for bidirectional sites. Wind-rose-dependent buffer standards could allow denser development at low-risk sites (bidirectional, e.g., DEI) while maintaining protection at high-risk sites (unidirectional, e.g., German Bight). This is directly relevant to the emerging cross-border coordination frameworks for the North Sea (Fliegner et al., 2025).

3.4 The ambush effect: directional vulnerability

Figure 6 presents regret cross-sections at $K = 500$ (converged) for four elliptical wind roses spanning the shape space. The K=500 results show peak regret ratios of 1.07–1.11$\times$ those at K=100, confirming that the directional pattern is robust while magnitudes increase slightly with more optimization starts. Peak bearings are unchanged from K=100 to K=500 for three cases and shift slightly (from 135 ° to 105 °) for the mid-range case (Table 2).

Peak regret consistently occurs at oblique angles to the dominant wind (270 °), not along it. We term this the "ambush effect": the liberal layout is pre-adapted to the dominant direction and therefore partially resilient to aligned neighbors. Off-axis neighbors exploit the layout's blind spots—directions the liberal optimizer did not prioritize.

Table 2. Peak regret from cross-sections at $K = 500$ (converged). Dominant wind: 270 °.

Case	Max (GWh)	Max (%)	Bearing	Distance
$a = 0.5$, $f = 1.0$	174.1	4.42	135 °	$5D$
DEI real wind rose	143.6	2.59	225 °	$15D$
$a = 0.9$, $f = 1.0$	128.6	3.24	315 °	$15D$
$a = 0.7$, $f = 0.5$	127.0	3.21	105 °	$15D$
$a = 0.9$, $f = 0.0$	105.9	2.67	195 °	$60D$
$a = 0.5$, $f = 0.0$	95.3	2.42	90 °	$15D$

For North Sea cluster planning, this finding is counterintuitive: a developer might assume the most dangerous neighbor is the one directly upwind. The ambush effect shows that neighbors at oblique angles—perhaps in an adjacent country's lease area at a different bearing—may cause greater design regret. This is relevant for cross-border coordination in the North Sea, where multiple national lease areas surround each farm at various bearings.

All six cross-sections are fully converged at $K = 500$. The DEI real wind rose shows peak regret of 143.6 GWh (2.59 %) at bearing 225 ° and $15D$—consistent with the "ambush effect," as the DEI prevailing wind is from the southwest ($\theta_{\text{prev}} \approx 251°$) and peak regret occurs 26 ° south of the dominant direction.

3.5 Self-interested neighbor comparison

The cross-section analyses above use a fixed reference farm translated to each position. To test whether the ambush effect persists under realistic neighbor behavior, we additionally optimize the reference farm to maximize its *own* AEP (given the target at its liberal layout) before re-optimizing the target farm. This self-interested variant represents a neighbor that acts rationally for its own production, not adversarially against the target.

Figure 7 compares the adversarial and self-interested cross-sections at $K = 500$ for three wind roses spanning the shape space. The results reveal a striking wind-rose dependence:

- Under bidirectional conditions ($a = 0.5$, $f = 0.0$), the self-interested neighbor produces 60 % of the adversarial peak regret (1.45 % vs. 2.42 % of AEP). The adversarial upper bound substantially overstates the realistic risk.
- Under mid-range conditions ($a = 0.7$, $f = 0.5$), a similar reduction occurs: 63 % of adversarial (2.01 % vs. 3.21 %).
- Under concentrated unidirectional conditions ($a = 0.9$, $f = 1.0$), the self-interested neighbor produces *nearly double* the adversarial peak regret: 5.99 % vs. 3.24 % of AEP.

The last result is the most policy-relevant: when wind is strongly unidirectional, a neighbor optimizing its own AEP concentrates its turbines upwind—exactly where the strongest wind is, and exactly where they cause the most damage to the target. The fixed reference grid, by contrast, spreads turbines evenly across its footprint, diluting the wake impact. This

Table 3. Fitted elliptical wind rose parameters for North Sea offshore sites. Data sources: Horns Rev 1 and Lillgrund from PyWake; Borssele from IEA Wind Task 55 ROWP; Princess Amalia from the python-windrose package; DEI from 10-year NEWA timeseries.

Site	a	f	θ_{prev}	R^2
Horns Rev 1 (DK)	0.649	0.384	240 °	0.82
Lillgrund (SE)	0.621	0.544	236 °	0.77
Borssele (NL)	0.728	0.291	240 °	0.87
Princess Amalia (NL)	0.650	0.182	210 °	0.49
Danish Energy Island	0.637	0.384	251 °	0.84

means that for concentrated unidirectional sites, the adversarial upper bound *underestimates* the design regret from a rational neighbor.

For North Sea planning, this implies that sites in the German Bight with strongly unidirectional southwesterly winds face *higher* design regret from realistic neighbor behavior than from adversarial placement—the "wind theft" problem is worse than the worst case suggests.

4 Discussion

4.1 Real North Sea sites in the shape space

To ground the parametric results in real planning decisions, we fit the elliptical model to five North Sea offshore wind sites (Table 3, Fig. 3). All five sites fall in the moderate-regret region of the (a, f) space, with $a \in [0.62, 0.73]$ and $f \in [0.18, 0.54]$. No operational North Sea site in our sample reaches the high-regret corner ($a > 0.8$, $f > 0.75$), though this does not preclude future lease areas—particularly in the southern North Sea—from doing so.

The sites cluster in the (a, f) space because North Sea wind climates share a common driver: Atlantic weather systems producing dominant southwesterly winds. The spread in f (0.18–0.54) reflects varying degrees of bidirectionality, with Princess Amalia showing the most bidirectional character ($f = 0.18$, lowest regret risk) and Lillgrund the most unidirectional ($f = 0.54$, highest regret risk among the five).

4.2 Implications for North Sea cluster development

The six-fold variation in design regret across wind rose shape has direct implications for the North Sea's emerging cluster framework.

4.2.1 Economic magnitude.

For a representative 750 MW farm (50×15 MW), design regret of 2.5 % corresponds to ∼100 GWh/yr. At a contract-for-difference strike price of 70 EUR/MWh, this amounts to ∼7 M EUR/yr, or ∼175 M EUR over a 25-year operational lifetime.

Even at the moderate regret levels observed for real North Sea sites ($\sim$1 %), the lifetime cost exceeds 70 M EUR—comparable to the cost of a single turbine foundation. By contrast, the marginal cost of running a neighbor-aware layout optimization is negligible relative to project CAPEX.

4.2.2 German Bight and southern North Sea.

Sites with strongly unidirectional southwesterly winds sit in the high-regret region ($f \gtrsim 0.75$). Here, the value of coordinated design is highest, and buffer distances of 30–$40D$ (7–10 km) may be needed to reduce regret below 1 %. The projected 30 % capacity factor reduction from cluster wakes (Fliegner et al., 2025) is partially a design problem, not just a physics problem: some of this loss could be mitigated by neighbor-aware layout optimization.

4.2.3 Danish waters.

The DEI wind rose ($f \approx 0.38$) benefits from a natural resilience: wakes from any single direction are partially offset by energy captured from the opposing direction, limiting the rearrangement advantage of a neighbor-aware design. This explains the negligible design tradeoffs found by Quick et al. (2026). However, this resilience is a property of the wind climate, not the farm—other Danish sites with different wind characteristics may not share it.

4.2.4 Cross-border coordination.

The value of inter-farm coordination is itself wind-rose-dependent. Our results suggest that coordination frameworks like the "good neighbour" principle in the UK EN-3 guidance should be weighted by wind rose directionality. Fixed buffer distances—such as the NYSERDA 4 nmi recommendation—are adequate for all sites tested but inefficient: bidirectional sites need only $\sim 10D$ to achieve the same residual regret. Wind-rose-conditional buffers could allow denser development at low-risk sites without compromising high-risk ones.

4.3 The convergence problem

Our most broadly applicable finding is methodological. At $K = 5$, regret estimates can be off by nearly a factor of two. The cause is catastrophic cancellation: regret is a difference between two large, noisy AEP values. A 0.5 % miss in either optimization can produce a 50 % error in the difference.

This affects any study computing AEP differences between strategies. We recommend $K \geq 300$ for 50-turbine farms and proportionally more for larger farms. The computational cost at $K = 500$ was approximately 2 hours per wind rose on a single AMD MI250X GCD, or 70 GPU-hours for the full 35-case sweep.

4.4 Limitations

4.4.1 Wake model.

We use Bastankhah and Porté-Agel (2014) with $k = 0.04$ and root-sum-of-squares superposition. More sophisticated models (e.g., TurbOPark) may change absolute magnitudes. The qualitative findings—folding dominance, the ambush effect—depend on wake corridor geometry, not magnitude, and are expected to be robust.

4.4.2 Adversarial vs. self-interested.

The greedy grid search and fixed reference farm provide an adversarial framing. The self-interested comparison (Section 3) shows that the adversarial upper bound overestimates regret for bidirectional sites (60 % of adversarial) but *underestimates* it for concentrated unidirectional sites (1.85× adversarial). This asymmetry means the qualitative ordering of sites by risk is robust, but absolute magnitudes depend on neighbor behavior.

4.4.3 Constant wind speed.

Real North Sea sites have speed–direction correlations. If high-speed winds are concentrated in the dominant direction (common for storm-driven sites), regret could be amplified through the cubic power relationship.

4.4.4 Single farm geometry.

Results may depend on the DEI polygon shape. Different boundaries could affect the layout optimization landscape and the potential for lateral escape from wake corridors.

4.4.5 Cross-section resolution.

The cross-section analysis uses 24 bearings and 7 distances, which may miss fine angular structure in the regret field.

5 Conclusions

1. **Standard multistart practice drastically underestimates design regret.** Five random starts capture only 55–86 % of converged regret. All configurations stabilize by $K = 300$, with no change through $K = 2000$. Any study computing AEP differences between strategies should use $K \geq 300$.

2. **Design regret varies six-fold across the wind rose shape space.** At $K = 500$, regret ranges from 17 GWh/yr (0.4 %) under bidirectional conditions to 101 GWh/yr (2.5 %) under concentrated unidirectional conditions. Folding (f, bidirectional vs. unidirectional) is the primary driver.

3. **Buffer distance requirements are wind-rose-dependent.** Concentrated unidirectional sites retain 0.4 % residual regret at $40D$ (9.6 km). Bidirectional sites achieve comparable levels at $10D$ (2.4 km). Wind-rose-conditional buffer standards could allow denser North Sea development at low-risk sites.

4. **Peak vulnerability occurs at oblique angles to the dominant wind.** The "ambush effect" means that off-axis neighbors—perhaps from adjacent countries' lease areas—may cause greater design regret than directly upwind neighbors. This is relevant for cross-border coordination in the North Sea.

The Danish Energy Island, with its partially bidirectional wind rose ($f \approx 0.38$), sits in the low-to-moderate regret region. Sites in the southern North Sea with strongly unidirectional southwesterly winds are substantially more vulnerable. As North Sea clustering intensifies toward the 300 GW target, incorporating wind rose directionality into buffer distance standards and coordination frameworks will become increasingly important for efficient maritime spatial planning.

. The simulation framework (pixwake) and analysis scripts are available at https://github.com/kilojoules/cluster-tradeoffs. The elliptical wind rose parameterization uses the edrose package (Hart, 2025), available at https://github.com/kilojoules/edrose.

. JQ designed the study, developed the simulation framework, performed all computations, and wrote the manuscript. PER supervised the research and contributed to methodology and manuscript revision.

. The authors declare no competing interests.

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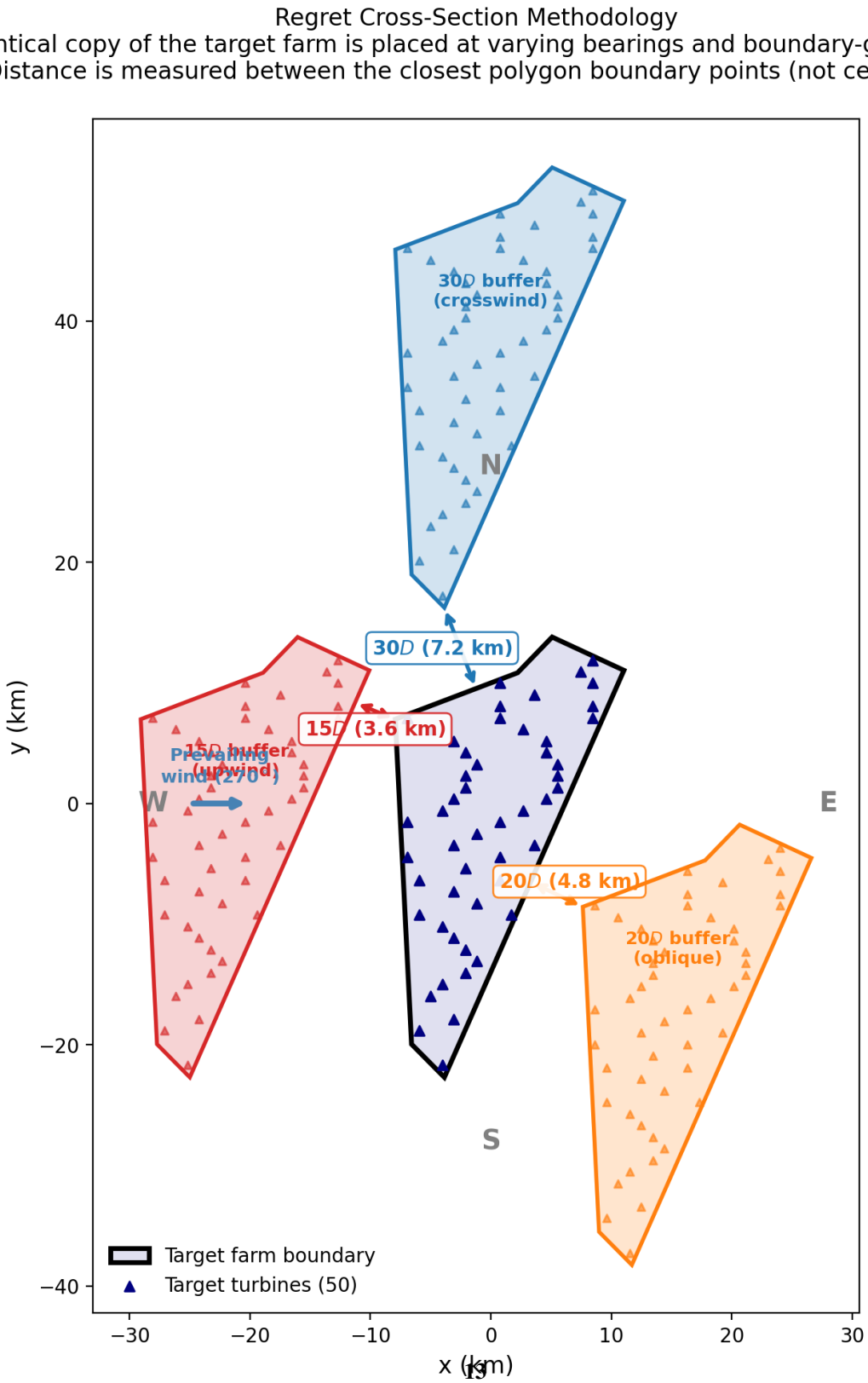

Figure 1. Regret cross-section methodology. Left: the target farm (gray polygon with blue turbines) surrounded by example reference farm

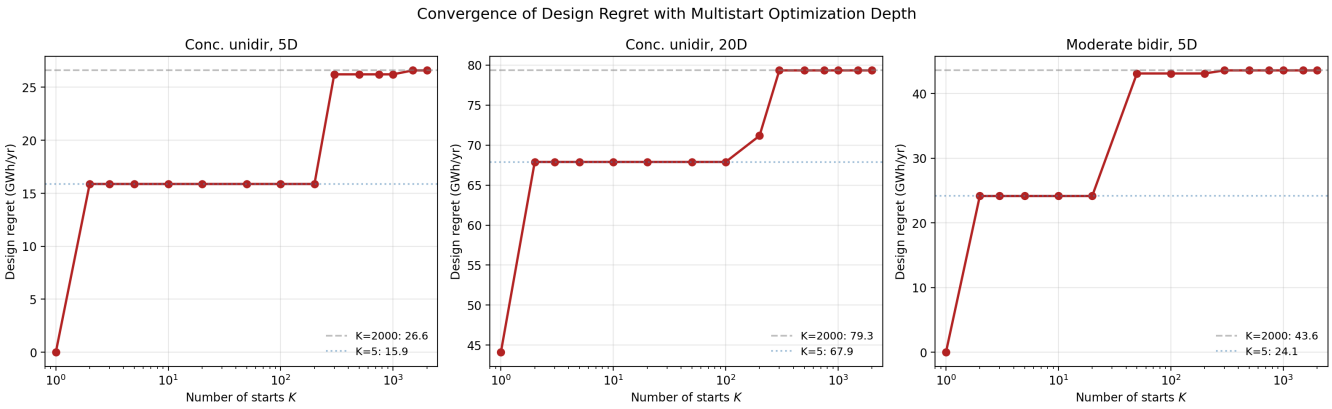

Figure 2. Convergence of design regret with multistart depth K. Dashed gray: converged value ($K = 2000$). Dotted blue: $K = 5$. All configurations stabilize by $K \approx 300$.

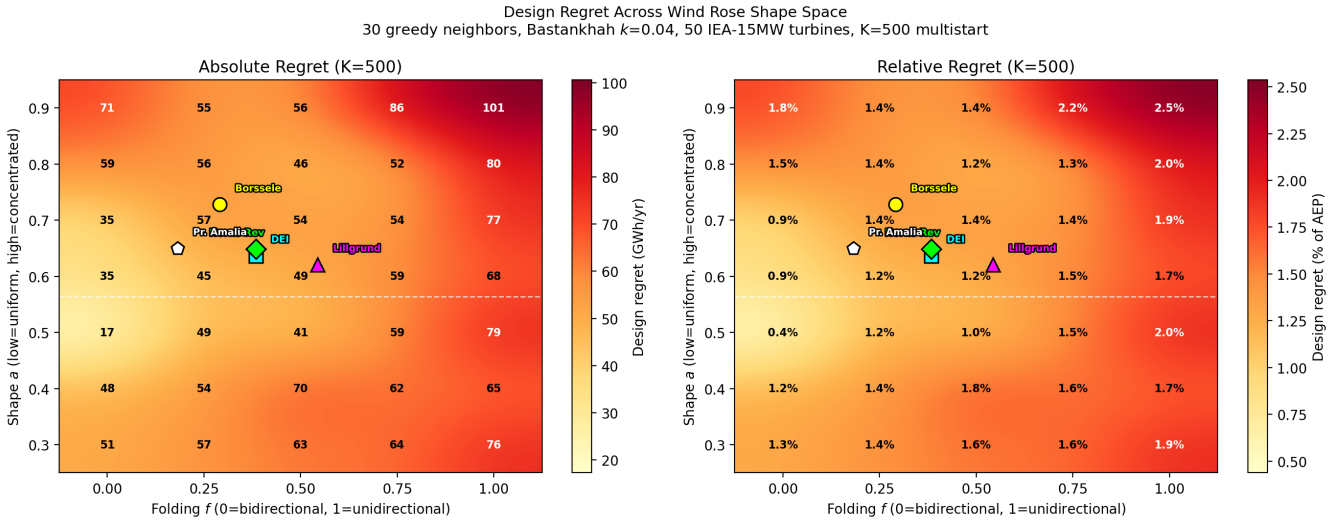

Figure 3. Design regret across the wind rose shape space at $K = 500$. Left: absolute (GWh/yr). Right: relative (% of AEP). Markers show fitted parameters for real North Sea sites: DEI (cyan square), Horns Rev 1 (green diamond), Lillgrund (magenta triangle), Borssele (yellow circle), and Princess Amalia (white pentagon). All real sites fall in the moderate-regret region of the shape space.

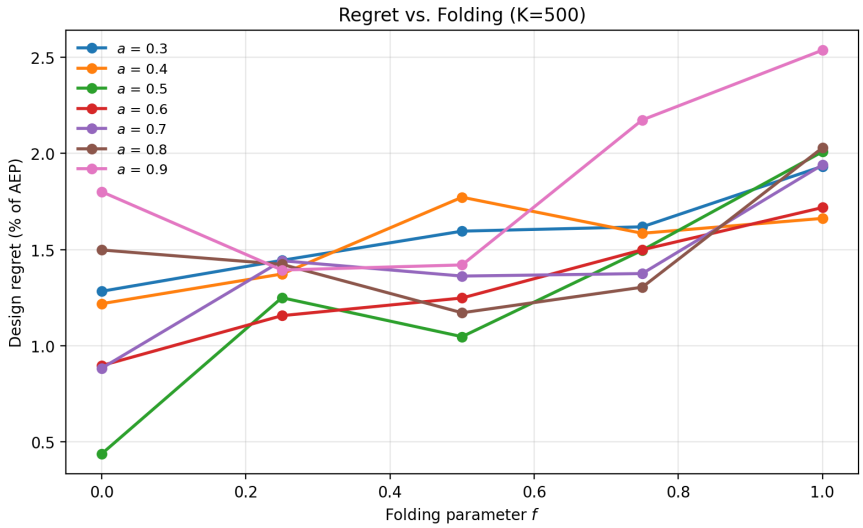

Figure 4. Regret vs. folding f at each concentration a ($K = 500$). Monotonic increase with f is consistent across all a.

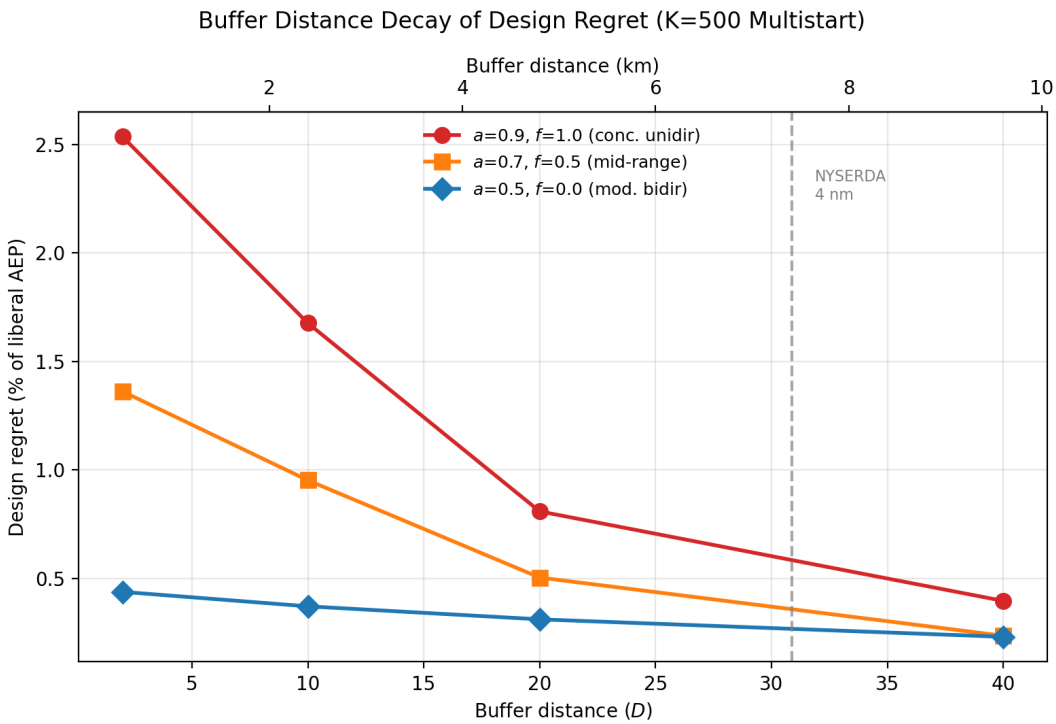

Figure 5. Regret vs. buffer distance at $K = 500$, expressed as % of liberal AEP. The NYSERDA 4 nmi recommendation is marked.

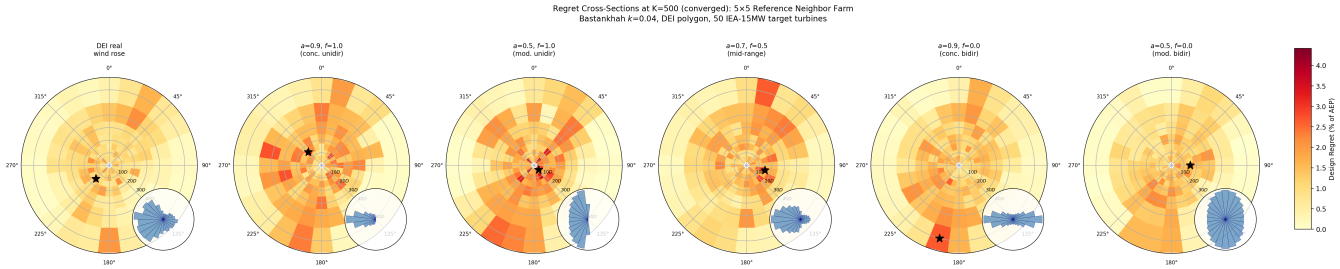

Figure 6. Regret cross-sections at $K = 500$: design regret vs. neighbor bearing and distance for a 5×5 reference farm. Stars mark peak regret. Inset wind roses show sector frequencies. Dominant wind: 270° (west).

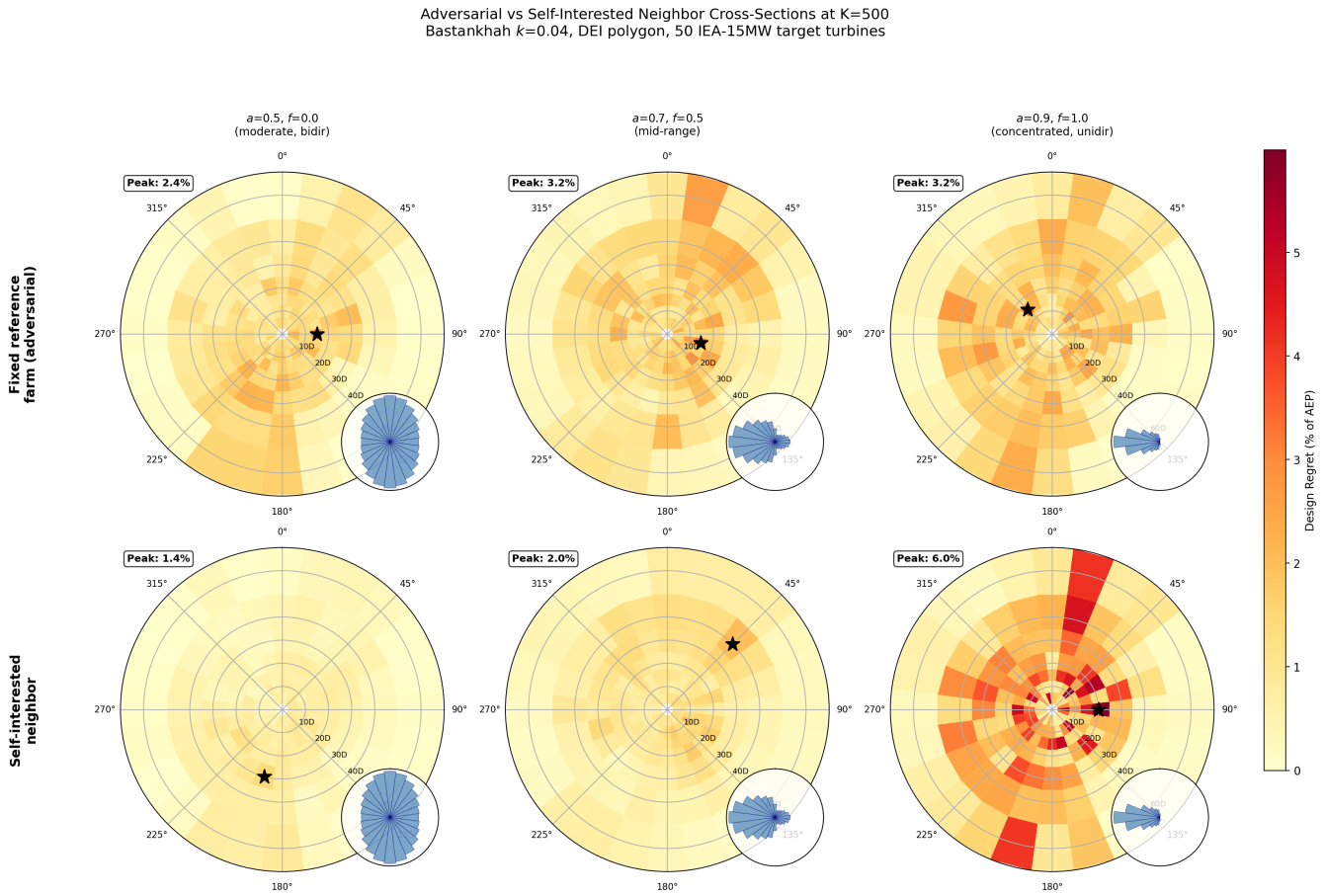

Figure 7. Adversarial (top row) vs. self-interested (bottom row) regret cross-sections at $K = 500$ for three wind roses. Under concentrated unidirectional wind ($a = 0.9$, $f = 1.0$), the self-interested neighbor produces nearly double the adversarial peak regret because AEP-optimal neighbor placement concentrates turbines in the high-wind upwind corridor.